

Ammar Nawawi

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PROFESSIONAL SUMMARY

Process and data engineer with 6+ years in semiconductor manufacturing — specialising in Statistical Process Control, process qualification, data-driven optimisation, and continuous improvement across high-volume production lines. Built and deployed production-grade SPC and MSA analytics platforms from scratch in Python, demonstrating technical depth beyond standard tooling: implementing statistical algorithms (ANOVA, variance components, Cpk, WECO detection) from first principles and sustaining them in daily production use. Experienced in end-to-end DMAIC projects, NPI process development, multivariate statistical analysis, and cross-functional team collaboration — with the full-stack capability to design, build, and own data systems that serve production teams directly.

CORE COMPETENCIES

Statistical Process Control (SPC / FDC)

Process Capability & MSA (Cpk, GR&R)

DOE (Full / Fractional Factorial)

Statistical Analysis (ANOVA, Regression)

Root Cause Analysis (8D, Fishbone)

Production Platform Development (Python)

Continuous Improvement (DMAIC / Lean)

NPI Process Development & Launch

FMEA / PFMEA & Process Mapping

Data Analytics (Python · SQL · JMP)

Cross-functional Collaboration

Training & Mentoring

WORK EXPERIENCE

TF-AMD Microelectronics (M) Sdn. Bhd.

Nov 2022 – Present

Process Engineer · Bayan Lepas, Penang

SPC Engineer (Mar 2024 – Present) → Visual Mechanical / ICOS Inspection (Nov 2022 – Mar 2024)

- Built and deployed a production-grade process monitoring platform from scratch** (Python/FastAPI/PostgreSQL) — replaced manual monitoring workflows entirely; used daily by 15+ floor engineers across 4+ active product lines with zero external tooling cost.
- Monitored production lines via SPC/FDC metrics** — maintained control charts, computed Cpk/process capability, detected WECO rule violations, and triaged out-of-control signals before yield impact across BGA, PGA, and LGA product families.
- Applied multivariate statistical analysis** (DOE, ANOVA, regression) to isolate root causes of process variation — separating true process shifts from measurement noise to drive reliable, data-backed decisions.
- Led multiple end-to-end DMAIC projects** — defined scope, collected production data, isolated root causes, and implemented verified corrective actions with documented, measurable outcomes across 3 product families on 4+ active lines.
- Trained 10+ Process and PDE Engineers** on SPC, measurement system analysis, and data-driven problem solving — built sustained analytical capability across the production floor.

ASE Electronics (M) Sdn. Bhd.

Jan 2022 – Nov 2022

Process Engineer

- Designed and optimised die attach process parameters via structured DOE** — mapped parameter interactions using full factorial experiments, identified optimal settings, and validated process capability with statistical confidence across SD card, micro SD, and USB connector product lines.
- Used **JMP for statistical analysis** — correlation, regression, and hypothesis testing — to identify and rank top yield loss contributors and guide process decisions.
- Resolved recurring die crack and equipment failures through **8D root cause analysis**, delivering evidence-backed corrective actions with sustained results.

Process Development Engineer

- **Led NPI process development for communication product lines** — designed operation sequences, developed PFMEA, created work instructions, tooling recommendations, and measurement plans for new assembly processes from introduction through production release.
- Developed **capacity models and cycle time standards** for new assembly lines — applied line balancing and time study methods to establish throughput baselines and identify bottlenecks before volume ramp.

PERSONAL PROJECTS

SPC Process Monitoring Platform LIVE · Production

Built a production-grade process monitoring system from scratch and deployed it at TF-AMD — computes Cpk, control limits, and WECO violations in real time with live chart rendering. Used daily by 15+ floor engineers across 4+ active product lines. Self-initiated: identified the monitoring gap, designed the solution, implemented it end-to-end, and sustained it in production.

Stack: Python · FastAPI · PostgreSQL · Vanilla JS · Chart.js

MSA Analytics Platform LIVE · Production

Web-based measurement system analysis tool computing the complete AIAG GR&R output — variance components, repeatability, reproducibility, P/T ratio, ndc, ANOVA table with F-statistics, and all standard MSA charts. Implemented from first principles in Python, validated against AIAG 4th Edition reference values and Minitab. Replaced Minitab for routine gauge studies across production.

Stack: Python · FastAPI · PostgreSQL · Vanilla JS · Chart.js

EDUCATION

BSc. Mechanical Engineering — Purdue University, West Lafayette, IN, USA

2014 – 2018

Dean's List (Jan–May 2015) · Coursework: System Controls, Design of Experiment, System Model & Analysis, Machine Design, Engineering Economics

TECHNICAL SKILLS

Process & Manufacturing Engineering: SPC (Xbar-R, EWMA, CUSUM, WECO), process qualification, capability studies (Cpk/Ppk), MSA (GR&R, AIAG 4th Ed.), FMEA/PFMEA, process mapping, NPI process development, operation sequence design, work instruction development, DOE (full/fractional factorial), ANOVA, regression, root cause analysis (8D, fishbone), DMAIC/Lean, line balancing, capacity modelling

Data, Analytics & Development: Python (data analysis, statistical computation, automation, web backend), FastAPI, SQL/PostgreSQL, Vanilla JavaScript, Chart.js, JMP, Minitab, MATLAB, Microsoft Excel (advanced)

Domain: Semiconductor backend manufacturing (TF-AMD / ASE Electronics), electronics assembly (Sanmina), ICOS automated optical inspection, BGA/PGA/LGA product families

Languages: English (fluent), Malay (native)